



Spring 2026

EMCH792 – Optimal State Estimation

Homework 2

Submission deadline: Thursday, 02/12/2026, 11:59pm EST

- Submission only through Blackboard, email submissions will not be accepted.
- Make sure you properly upload the file(s) and that you can successfully view your submission on BB. Keep a copy of the submission receipt you get from BB.
- Late submissions will be graded with the policy described in the syllabus.
- For questions email tsucin@email.sc.edu and copy vitzilaos@sc.edu

Homework Instructions

In this homework, you must analyze linear systems using state-space models. All the files that you will need for this homework are in the **ws** folder. You are expected to work inside this folder only. Open **MATLAB** and set the current folder to **ws**, and do the following:

1. Start by opening the live script file "HW 2 manual.mlx" using **MATLAB**. This will guide you through basic **MATLAB** functions to analyze linear systems. Read through the file and understand how each function is implemented in **MATLAB**.
2. Open the file "HW 2.m" and follow the instructions below and, in the file, to solve all the questions of the problem.

Problem – The coordinated turn model

Consider an object moving in a 2D plane with constant speed and angular rate Ω where $\Omega > 0$ is counterclockwise motion and $\Omega < 0$ is clockwise motion. The system is described by the equations:

$$\begin{aligned}\ddot{\xi} &= -\Omega\dot{\eta} + u_1 \\ \ddot{\eta} &= \Omega\dot{\xi} + u_2\end{aligned}$$

The state and input of the system are:

$$x = \begin{bmatrix} \xi \\ \dot{\xi} \\ \eta \\ \dot{\eta} \end{bmatrix}, \quad u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

We can measure ξ, η .

For the described system do:

1. For $\Omega = 0.5$ define the continuous time system.
2. Check the system's controllability and observability.
3. Simulate the continuous time system in Simulink.
4. Find the discrete time system for a sampling time of $T = 0.1\text{sec}$.
5. Simulate the discrete time system in Simulink.
6. Simulate the discrete time system in MATLAB for 200 simulation steps.

BONUS: Find the discrete time system dynamics for any Ω, T . You can use **MATLAB** and symbolics.

Homework Submission

Your submission must include the following files:

- “HW_2.m” -> The edited MATLAB script with your solutions
- “Continuous_sim.slx” -> The Simulink file for the continuous time system
- “Discrete_sim.slx” -> The Simulink file for the discrete time system

Upload all files on Blackboard in a single submission.

Make sure that the scripts you submit can run. Don't copy the results that appear in the command window in your scripts and make sure that if you add any comments in your scripts that they follow the proper **MATLAB** syntax. **Scripts that do not run will not be graded.**

Grading Rubric

Problem	Points
1	15
2	15
3	20
4	10
5	20
6	20
BONUS ¹	15

¹ The bonus problem is optional and can help increase your score if you miss points in the other problems. Maximum score for the assignment is 100 points.